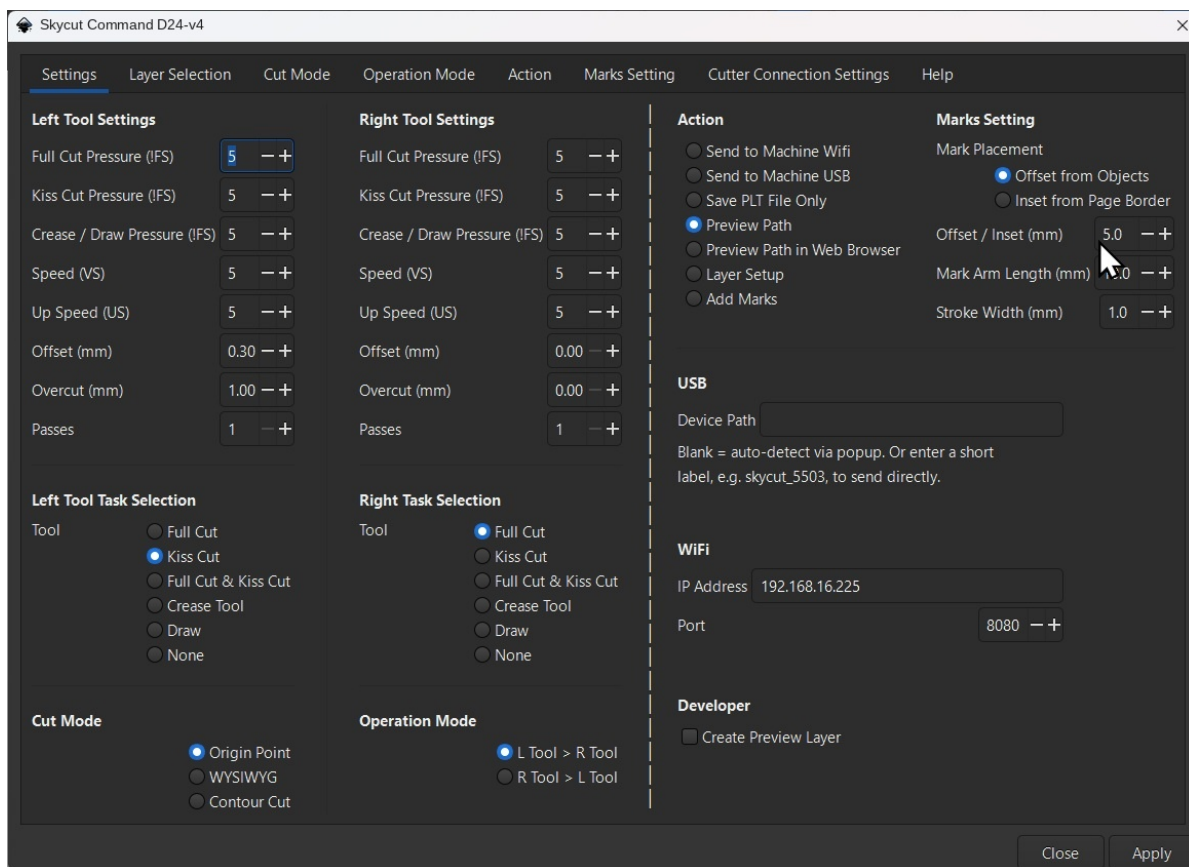


I'm not a code writer — much of this was created using AI. Also acknowledging trankata/inkscape-skycut-d24 extension as the inspiration, though this was taken in a whole other direction.

Skycut Command D24-v4

An Inkscape 1.x extension for driving **SkyCut** vinyl/sign cutters directly from Inkscape — Full Cut, Kiss Cut, Crease, and Draw passes, dual independent tool heads, knife-offset corner compensation, three coordinate-reference modes (including ContourCut registration-mark alignment), a built-in path previewer, and delivery to the cutter over WiFi, USB, or as a saved .plt file.

This is a **standalone** extension — all of its module files are suffixed `_d24_3` and it shares nothing with any earlier SkyCut extension, so multiple versions can be installed in Inkscape's extensions folder side by side without collisions.



Features

- **Dual independent tool heads.** Left and Right each get their own speed, up-speed, offset, overcut, passes, and source layer assignment. Either head can be set to Full Cut, Kiss Cut, **Full Cut & Kiss Cut on the same head**, Crease, Draw, or None.
- **Per-operation pressure.** Each side carries three separate pressure settings — Full Cut, Kiss Cut, and Crease/Draw — since one physical head running a combined Full Cut + Kiss Cut job may need different blade pressure for each pass even though speed/offset/overcut are shared.

- **Knife-offset corner compensation.** A swiveling drag-knife trails behind its own pivot point, so straight-line offset isn't enough at corners. `skycut_pathbuild_d24_3.py` inserts direction-aware overshoot/return loops (with a small outward Bezier bow to let the blade re-align) at every sharp corner, including the seam of a closed shape, plus a configurable overcut past the closing point.
- **Three coordinate-reference modes:**
 - **Origin Point** — reference box is just the bounding box of the artwork being cut.
 - **WYSIWYG** — reference box is the Inkscape page itself, preserving artwork position on the page.
 - **Contour Cut** — reference box comes from four L-shaped registration marks (see *Add Marks* below), for aligning a cut to pre-printed artwork.
- **Add Marks.** Generates registration marks (with an orientation indicator) on a fresh, locked Marks layer, either offset from the artwork's bounding box or inset from the page border.
- **Layer Setup.** One click creates (or refreshes the color/style of) the standard SkyCut layer set: Marks, Print, Full Cut, Kiss Cut, Crease, Draw.
- **Preview Path / Preview Path in Web Browser.** QC-only render of the exact toolpath that would be sent — corner overshoots, overcuts, and per-tool coloring included — shown in Inkscape's natural (un-rotated) orientation rather than the cutter's swapped-axis machine orientation.
- **Three delivery methods:** send over WiFi (raw TCP), send over USB (Windows, USB-Printer class device), or save a timestamped `.plt` file to `~/Documents/Skycut Data`.
- **Compound-path aware.** Each subpath of a compound `<path>` (e.g. a ring's outer contour plus an inner hole) is treated as its own independent contour with its own pen-lift, instead of being cut straight through into the next.
- **Automatic shape-to-path conversion.** Rectangles, circles, ellipses, polygons, lines, and live text objects drawn with Inkscape's shape/text tools aren't `<svg:path>` elements, so they'd otherwise be silently skipped when cutting — no error, no visual difference on the canvas. Every run scans all four source layers for these and converts them to paths automatically (the same result as **Path** → **Object to Path / Text** → **Object to Path**), then stops so you can check the result on the canvas before sending the job.

Requirements

- Inkscape 1.x (uses the modern `inkex` API — `Layer.new`, `svg.get_page_bbox()`, etc.)
- Python 3 (bundled with Inkscape)
- **Send to Machine USB** only works on Windows (uses `ctypes` + `SetupAPI/kernel32` to talk to the USB-Printer class device, VID 0483 / PID 5750). WiFi send and Save PLT File work on any platform Inkscape supports.
- The USB device picker window requires `tkinter`; if it isn't available and more than one matching device is found, the extension will list device labels for you to paste into the USB Device Path field instead.

Installation

1. Copy every file in this repo into your Inkscape user extensions folder:

- Windows: `%APPDATA%\inkscape\extensions\`

2. Restart Inkscape.

3. The extension appears under **Extensions** → **Skycut Command D24-v4**.

File overview

File	Purpose
skycut_command_d24_3.inx	Inkscape UI definition (the dialog's tabs, fields, and menu entry)
skycut_command_d24_3.py	Entry point — reads options, orchestrates the pipeline, dispatches to Send/Save/Preview
skycut_constants_d24_3.py	Shared master switches and numeric constants (scale, colors, layer names, etc.)
skycut_geometry_d24_3.py	Pure 2D vector math + cubic-Bezier flattening (no inkex dependency)
skycut_pathbuild_d24_3.py	Knife-offset corner overshoot/return generation and final overcut
skycut_reference_d24_3.py	Origin / WYSIWYG / ContourCut reference-box computation
skycut_layers_d24_3.py	SVG layer management, geometry collection, and debug/preview capture (mixin)
skycut_layer_setup_d24_3.py	Creates/refreshes the standard SkyCut layer set
skycut_add_marks_d24_3.py	Generates ContourCut registration marks
skycut_arrows_d24_3.py	Debug-only cut-direction arrowhead helper
skycut_hpgl_d24_3.py	Assembles the final HPGL command stream (and the un-swapped viewer render segments)
skycut_output_d24_3.py	Delivers HPGL to disk, WiFi (TCP), or USB
skycut_usb_d24_3.py	Windows USB-Printer-class device discovery and raw send
skycut_viewer_d24_3.py	Preview Path viewer

Usage

1. Set up your artwork in Inkscape on layers named **Full Cut**, **Kiss Cut**, **Crease**, and/or **Draw** — the layer names must match (case-insensitively), or use **Output** → **Layer Setup** to have the extension create them for you with the standard colors.
2. Open **Extensions** → **Skycut Command D24-v4**.
3. On the **Left Tool** / **Right Tool** tabs, choose which layer each physical head should cut and set its speed, offset, overcut, and pressure(s).
4. Choose a **Cut Mode**:
 - **Origin Point** for artwork-only reference.
 - **WYSIWYG** to preserve the artwork's position on the physical page.
 - **Contour Cut** to align to registration marks (run **Output** → **Add Marks** first, print the design, then re-scan the marks before cutting).
5. Under **Action**, pick an output:
 - **Preview Path** / **Preview Path in Web Browser** to check the toolpath first — no data is sent or saved.

■ **Send to Machine Wifi / Send to Machine USB** to cut immediately.

■ **Save PLT File Only** to write a .plt file for later.

6. Run the extension.

A note on coordinates

Inkscape uses a top-left-origin, Y-down page. SkyCut's HPGL machine coordinates are bottom-left-origin, Y-up, **with X and Y swapped** relative to normal HPGL ($hpgl_x = max_y - y$, $hpgl_y = max_x - x$). All of that conversion happens once, right before HPGL emission — the Preview Path viewer deliberately skips the axis swap so what you see on screen matches your artwork's natural orientation in Inkscape, while still reflecting the exact same offsets, overcuts, and corner handling as the real output.

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